AIT

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Project Title: Bus Seat Booking System

Group members:

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Submitted to:

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- Project Objective:

The objective of this project is to develop a reliable and user-friendly Bus Seat Booking System that streamlines the process of reserving seats on buses. It aims to eliminate manual booking errors, provide real-time seat availability, and offer administrative control over routes, schedules, and pricing. The system is designed to enhance operational efficiency and improve the user experience for both passengers and administrators.

- Project Abstraction:

This web-based application consists of two primary modules: the User Side and the Admin Side. Users can view available routes, check seat availability, and make or cancel bookings. The Admin Side provides full control over system configurations, including route management, bus scheduling, seat layout, and pricing. The system uses Java (Servlets and JSP) for backend logic, MySQL for data storage, and Apache Tomcat as the server environment. Real-time updates and a simple login system

ensure smooth interaction and secure access for all users. ----------

- **Project Conclusion:**

The Bus Seat Booking System successfully demonstrates the integration of frontend and backend technologies to solve a real-world problem in transportation management. It provides a scalable and efficient solution for booking and managing bus seats, with clear benefits for both users and administrators. Through this project, the team gained hands-on experience in full-stack development, database connectivity, and system design—laying a strong foundation for future professional work in web development.

Further Details

- Project Overview:

This project is a web-based Bus Seat Booking System designed for educational purposes. It allows users to book seats on buses and provides an admin panel for managing routes, schedules, buses, and pricing. The system is divided into two main modules: User Side and Admin Side.

- System Roles:

1. User Side:

- View available routes and timings

- View buses and seat availability

- View fixed pricing before booking

- Book a seat

- Cancel a booking

- User necessary Information (Required in case of any problem)

- View charges and booking history

- Help contact number of Bus organization

2. Admin Side:

- Add/edit/delete routes
- Set bus schedules and timings
- Add/remove buses
- Manage seat layouts and availability
- Set fixed pricing before bookings are allowed
- View all bookings
- Cancel or book seats manually
- Get access the stored information of users or customers
- Set the Information about your organization and contact number

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⭘ Key Features:

- Fixed pricing must be set and changed by admin
- Seat availability updates in real time after booking/cancellation
- Simple and secure login system for both users and admin
- Admin dashboard for full control over system settings

⭘ Technologies to Learn & Use:

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Java (Servlets and JSP)

Database:

- MySQL (via JDBC)

Server:

- Apache Tomcat

⭘ Folder Structure:

```
BusBookingSystem/
├── docs/
|   └── student_plan.pdf
├── src/
|   ├── UserServlet.java
|   ├── AdminServlet.java
|   └── PricingManager.java
├── web/
|   ├── index.jsp
|   ├── booking.jsp
|   ├── admin_panel.jsp
|   └── pricing.jsp
└── README.md
```

⭘ Development Phases:

Phase 1: Setup and Environment Configuration

Phase 2: Design User Interface (HTML/CSS)

Phase 3: Implement Login System

Phase 4: Admin Panel (Routes, Buses, Pricing)

Phase 5: Booking Logic and Seat Management

Phase 6: Testing and Debugging

Phase 7: Final Presentation and Documentation